

# Lachlan Wright

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## About Me

Full-stack Software Engineer with experience developing and deploying web applications. Looking to deepen my experience and capabilities while expanding my skillset across new frameworks and languages.

## Experience

### Software Engineer - Chaos1 (Sydney, NSW)

Nov 2023 - Feb 2026

- Developed a NextJS website with TailwindCSS and ShadCN with a FastAPI backend.
- Contributed to a multi-user text editor utilising TipTap and HocusPocus.
- Developed an additional Flutter application with a Spring backend utilising the Hibernate ORM.
- Managed Google Cloud Platform services, including Cloud Build, Cloud Run, and Cloud SQL.
- Implemented continuous integration testing suites executed using Pytest, Junit, and Cypress.
- Provided team members with technical mentoring, guidance, and code reviews.

### Programmer (Freelancer) - DataAnnotation (Remote)

Jul 2024 - Present

- Wrote and reviewed programming prompts and responses to assist the development of LLMs.

## Education & Qualifications

### Bachelor of Engineering (Honours) (Software) - UTS

Feb 2020 - Nov 2024

- Completed degree with 198 credit points, with a WAM of 85.97%, and a GPA of 6.47/7.00.

### Microsoft Certified: Azure Fundamentals

Jul 2021

- Demonstrated foundational knowledge of cloud services and how they are provided with Azure.

## Portfolio

### Pangea Resource Edit

- Formulated a level editor and model viewer for 8 games with React, Konva, and ThreeJS.
- Rewrote and extended a Python library for editing MacOS Resource Fork files in TypeScript.
- Implemented the LZSS compression algorithm for decompressing and repacking level data.
- Developed converter for translating model data from a proprietary format to glTF and back.
- Developed WebAssembly ports of games for in-browser testing of levels.

### Online Chat Game

- Constructed an online chat-based game built with NextJS, in addition to a React Native mobile app.
- Handled bidirectional user-server WebSocket communication using SocketIO and PartyKit.

### Driving Game and Level Editor

- Ported a MacOS game to WebAssembly, added Lua scripting and developed an Angular level editor.
- Produced Angular-based level and asset editors hosted alongside the game on GitHub Pages.

### Public Transport Timetable

- Developed a Flutter mobile application for viewing public transportation timetables in NSW.
- Interfaced with Transport for NSW's OpenData API, parsing Protobuf and CSV data.
- Persisted local data with Drift, and displayed geographical information with the Flutter Map library.

### Deal Scanner

- Devised a Go application for monitoring deals from online retailers using Playwright-based scraping.
- Persisted queries with GORM and automated deployment to Oracle Cloud with GitHub Actions.

### WiiTarThing Port

- Ported a Windows Presentation Foundation application to WinUI 3 utilising fluent design.
- Restructured the application to follow a MVVM-based architecture.

### Online Exam System

- Created a mock website for online invigilation of exams with NextJS, tRPC, Prisma, and TailwindCSS.
- Provided advice, feedback, guidance, and performed code reviews in an 8-person team.

## Technical & Core Capabilities

**Languages:** C, C#, Dart, Go, Java, JavaScript, Kotlin, MATLAB, Python, Rust, TypeScript, HTML/CSS

**Frameworks & Libraries:** Angular, Bootstrap, ExpressJS, Flutter, Immer, Jotai, KNIME, .NET, NextJS, NumPy, Pandas, PartyKit, Pydantic, Pyodide, React, React Native, ShadCN, Socket.IO, SolidJS, Spring, Svelte, SvelteKit, TailwindCSS, TanStack, TensorFlow, TipTap, tRPC, Vue, WinUI

**Databases:** Hibernate, MongoDB, Mongoose, PostgreSQL, Prisma, SQLAlchemy, SQLite

**CI/CD:** Azure, Cypress, Docker, GitHub Actions, GitLab CI/CD, GCP, Heroku, Playwright, Pytest, Vitest

**Agentic Tools:** Antigravity, Claude Code, Codex, Gemini CLI, GitHub Copilot, OpenCode